

Chapter 14C

Male Reproductive and Urologic Medicine

Medvale Internal Medicine - Reproductive Medicine Module

Scope and purpose. This segment covers male reproductive physiology, hypogonadism, erectile dysfunction, male infertility, acute scrotum, benign scrotal disease, male reproductive infections, benign prostatic hyperplasia, prostatitis, prostate cancer, testicular cancer, and high-yield diagnostic algorithms. The chapter is written as an original synthesis for study and Graph-RAG ingestion; outside guidelines and Anki signals were used as coverage anchors rather than copied text.

1. Core Male Reproductive Physiology

Male reproductive medicine is easiest to organize around three linked systems: the endocrine testis, the exocrine testis, and the outflow tract. The endocrine testis produces testosterone through Leydig cells under luteinizing hormone stimulation. The exocrine testis produces sperm through seminiferous tubules, Sertoli cells, intratesticular testosterone, and follicle-stimulating hormone. The outflow tract stores, matures, and transports sperm through the epididymis, vas deferens, ejaculatory ducts, prostate, seminal vesicles, and urethra.

GnRH pulses from the hypothalamus stimulate the anterior pituitary to release LH and FSH. LH stimulates Leydig cells to synthesize testosterone. FSH stimulates Sertoli cells to support spermatogenesis and produce inhibin B. Testosterone and estradiol provide negative feedback to the hypothalamus and pituitary, whereas inhibin B selectively suppresses FSH. A low testosterone level with high LH/FSH implies primary testicular failure. A low testosterone level with low or inappropriately normal LH/FSH implies secondary hypogonadism from hypothalamic, pituitary, medication, systemic illness, obesity, sleep apnea, or hyperprolactinemia-related suppression.

Structure / hormone	Main function	Clinical relationship
GnRH	Pulsatile hypothalamic signal for LH/FSH	Continuous GnRH suppresses gonadotropins; low pulsatility causes hypogonadotropic hypogonadism.
LH	Stimulates Leydig cells	Low LH causes low testosterone; high LH with low testosterone suggests primary testicular dysfunction.
FSH	Stimulates Sertoli cells and spermatogenesis	High FSH often marks seminiferous tubule injury; low FSH may impair sperm production.
Leydig cell	Testosterone synthesis	Damaged by primary testicular failure, chemo/radiation, orchitis, trauma, Klinefelter syndrome.
Sertoli cell	Supports germ cells; produces inhibin B	Failure causes oligospermia/azoospermia and often high FSH.
Epididymis	Sperm maturation and storage	Painful inflammation is epididymitis; obstruction can cause male infertility.
Prostate / seminal vesicles	Contribute seminal fluid	BPH causes outlet obstruction; prostatitis causes pelvic pain and urinary symptoms.

2. Initial Male Reproductive Symptom Framework

Male reproductive complaints often present as pain, mass, discharge, fertility difficulty, sexual dysfunction, urinary symptoms, or incidental laboratory abnormality. The first diagnostic step is not to name the most common disease but to exclude time-sensitive and malignant disease. Acute testicular pain is torsion until proven otherwise. A firm intratesticular mass is testicular cancer until proven otherwise. Fever, urethral discharge, dysuria, and epididymal tenderness suggest infection. Progressive lower urinary tract symptoms in an older man suggest BPH, but hematuria, bone pain, weight loss, renal failure, recurrent urinary retention, or abnormal prostate examination should broaden the differential.

Presentation	Do-not-miss diagnosis	First pattern recognition step
Sudden severe unilateral testicular pain	Testicular torsion	Assess time course, high-riding testis, absent cremasteric reflex; urgent urology if suspicious.

Presentation	Do-not-miss diagnosis	First pattern recognition step
Painless firm testicular mass	Testicular germ cell tumor	Scrotal ultrasound and serum tumor markers; do not biopsy through scrotum.
Testicular pain with dysuria/discharge	Epididymitis	NAAT for gonorrhea/chlamydia; urine studies; treat by age/sexual risk.
Penile discharge	Urethritis	Test for gonorrhea, chlamydia, HIV, syphilis when appropriate; treat partners.
Erectile dysfunction	ASCVD/diabetes/endocrine disease	ED may be an early vascular marker; check cardiometabolic and medication contributors.
Infertility	Male factor infertility	Obtain semen analysis early; evaluate both partners rather than assuming female factor.
Nocturia, weak stream, hesitancy	BPH or prostate cancer mimics	Assess severity, retention, renal dysfunction, infection, hematuria, medication contributors.
Gynecomastia	Malignancy or endocrine disease	Distinguish glandular enlargement from adipose tissue; review meds and testicular exam.

3. Hypogonadism and Testosterone Deficiency

Male hypogonadism is a clinical-biochemical syndrome. Symptoms alone are not sufficient because fatigue, low libido, erectile dysfunction, depressed mood, reduced muscle mass, anemia, and low bone density can have many causes. A low testosterone value alone is also not sufficient because testosterone varies by time of day, illness, sleep, obesity, medications, and assay factors. The practical approach is to confirm symptoms, measure an early morning total testosterone on two separate occasions, then determine whether the pattern is primary or secondary using LH and FSH.

The American Urological Association uses total testosterone below 300 ng/dL as a reasonable diagnostic cut-off when evaluating testosterone deficiency, but it emphasizes that the diagnosis also requires compatible signs or symptoms. This distinction is important for Graph-RAG: low testosterone is a laboratory node; testosterone deficiency is a clinical syndrome node that requires both a low hormone measurement and clinical context.

Pattern	Likely location	Examples	Expected labs
Low testosterone + high LH/FSH	Primary testicular failure	Klinefelter syndrome, orchitis, chemotherapy, radiation, trauma, aging-related testicular dysfunction	Low T, high LH, high FSH; semen abnormalities common if seminiferous tubules involved.
Low testosterone + low/normal LH/FSH	Hypothalamic-pituitary suppression	Obesity, opioids, glucocorticoids, hyperprolactinemia, pituitary tumor, systemic illness, sleep apnea	Low T with inappropriately low/normal gonadotropins; prolactin/iron/pituitary imaging depending on context.
Normal testosterone + infertility	Spermatogenic or obstructive disease	Varicocele, obstruction, genetic causes, prior vasectomy, heat/toxin exposure	Semen analysis abnormal; endocrine labs may be normal.
Gynecomastia + testicular mass	Estrogen or hCG-mediated tumor	Leydig/Sertoli tumor, germ cell tumor with hCG	Breast enlargement plus abnormal testicular findings or tumor markers.

Therapy should target the cause when possible. Weight loss, treatment of sleep apnea, reducing opioids or glucocorticoids when clinically appropriate, and correction of hyperprolactinemia may improve secondary hypogonadism. Exogenous testosterone can suppress LH and FSH, reduce intratesticular testosterone, impair spermatogenesis, and worsen infertility. Therefore, a man who desires future fertility should generally not be started on testosterone without a fertility-aware plan. Alternatives such as clomiphene, aromatase inhibition in selected cases, or gonadotropin therapy are specialist-managed approaches that preserve or stimulate the HPG axis rather than replacing it.

High-yield relationship: exogenous testosterone -> pituitary negative feedback -> low LH/FSH -> low intratesticular testosterone -> decreased spermatogenesis -> oligospermia/azoospermia -> infertility risk.

4. Erectile Dysfunction, Libido, and Ejaculatory Disorders

Erectile dysfunction is the persistent inability to obtain or maintain an erection sufficient for satisfactory sexual activity. The physiology requires arterial inflow, nitric oxide-mediated cavernosal smooth muscle relaxation, intact veno-occlusion, neurologic signaling, adequate testosterone for libido and erectile tissue health, and favorable psychological context. Because penile arteries are small, erectile dysfunction can precede symptomatic coronary artery disease and should trigger cardiovascular risk assessment, diabetes screening, medication review, and lifestyle counseling.

Mechanism	Clues	Common causes	Management concept
Vascular	Gradual onset, diabetes, smoking, HTN, PAD/CAD	Atherosclerosis, endothelial dysfunction, venous leak	Risk factor control; PDE5 inhibitor if safe; avoid nitrates.
Neurogenic	Diabetes neuropathy, spinal cord disease, pelvic surgery	Autonomic neuropathy, prostatectomy, neurologic disease	Treat cause; PDE5 inhibitor, vacuum device, injections, prosthesis if needed.
Endocrine	Low libido, infertility, fatigue, low morning T	Hypogonadism, hyperprolactinemia, thyroid disease	Confirm labs; treat endocrine disorder; fertility-aware testosterone decisions.
Medication/substance	Temporal relation to new drug	SSRIs, beta blockers, thiazides, opioids, alcohol, antiandrogens	Review alternatives; avoid abrupt stopping when medication is essential.
Psychogenic / relational	Situational ED, preserved nocturnal erections, performance anxiety	Anxiety, depression, stress, relationship conflict	Counseling, mental health care, sex therapy, PDE5 inhibitor as bridge.

PDE5 inhibitors enhance nitric oxide-cGMP signaling and are first-line for many patients. They are contraindicated with nitrates and require caution with alpha blockers, severe cardiovascular instability, and retinitis pigmentosa. Failure of a PDE5 inhibitor often results from incorrect use rather than true nonresponse: inadequate sexual stimulation, taking the medication after a heavy meal, insufficient dose trials, or unmanaged diabetes/hypogonadism can all reduce response.

Ejaculatory disorders include premature ejaculation, delayed ejaculation, anejaculation, and retrograde ejaculation. Retrograde ejaculation occurs when semen enters the bladder rather than exiting the urethra; it can follow diabetes autonomic neuropathy, prostate/bladder neck surgery, or alpha-blocker therapy. Post-ejaculatory urine containing sperm supports the diagnosis. Anejaculation with normal orgasm suggests neurologic, medication, or psychogenic causes; painful ejaculation raises concern for prostatitis, seminal vesicle disease, pelvic pain syndrome, or ejaculatory duct obstruction.

5. Male Infertility

Infertility is a couple-level diagnosis, but male factor abnormalities are common enough that semen analysis should occur early rather than only after an extensive female workup. The male evaluation identifies whether the problem is sperm production, sperm transport, sexual/ejaculatory function, endocrine signaling, genetic disease, or an environmental/toxin effect. AUA/ASRM guidance frames the male partner evaluation as a core part of infertile couple care rather than an optional late referral.

Category	Mechanism	Examples	Diagnostic clues
Pretesticular endocrine	Low gonadotropin drive or abnormal hormonal environment	Hypogonadotropic hypogonadism, hyperprolactinemia, thyroid disease, anabolic steroid/testosterone use	Low/normal LH/FSH with low T; low sperm count; history of opioids/anabolic agents.
Testicular sperm production failure	Seminiferous tubule or germ cell injury	Klinefelter syndrome, cryptorchidism, mumps orchitis, chemo/radiation, varicocele, heat/toxins	Small testes, high FSH, oligospermia/azoospermia.
Post-testicular obstruction	Sperm made but cannot exit	Vasectomy, congenital bilateral absence of vas deferens, ejaculatory duct obstruction, epididymal obstruction	Normal testicular size/hormones with azoospermia; low semen volume suggests ejaculatory duct/seminal vesicle issue.
Sexual/ejaculatory dysfunction	Semen not delivered effectively	ED, anejaculation, retrograde ejaculation, spinal cord injury, diabetic autonomic neuropathy	History identifies delivery problem; sperm may be found in post-ejaculatory urine.
Genetic	Chromosomal or Y-chromosome defects; CFTR-related obstruction	Klinefelter, Y microdeletion, congenital absence of vas deferens	Severe oligospermia/azoospermia; absent vas deferens; family/genetic clues.

Semen analysis measures volume, concentration, total sperm number, motility, morphology, and sometimes vitality. Abnormal results should generally be repeated because sperm parameters vary and spermatogenesis takes roughly 74 days. A low semen volume can suggest incomplete collection, retrograde ejaculation, androgen deficiency, ejaculatory duct obstruction, or congenital absence of seminal vesicles/vas deferens. Azoospermia requires separating obstructive from nonobstructive causes using testicular size, FSH, semen volume, pH/fructose when available, and specialist evaluation.

Semen pattern	Possible meaning	Next diagnostic direction
Low volume	Incomplete collection, retrograde ejaculation, ejaculatory duct obstruction, androgen deficiency	Repeat collection, post-ejaculatory urinalysis, consider obstruction if acidic/low fructose.
Oligospermia	Reduced sperm production or partial obstruction	History, testicular exam, FSH/LH/T, varicocele assessment.
Azoospermia + high FSH/small testes	Nonobstructive testicular failure	Genetic testing in severe cases; reproductive urology.
Azoospermia + normal testes/FSH	Obstruction or ejaculatory dysfunction	Check vas deferens, semen volume/pH, prior vasectomy/infection/surgery.
Asthenospermia	Poor motility	Varicocele, infection, antisperm antibodies, heat/toxins, lab artifact.
Teratospermia	Abnormal morphology	Nonspecific; interpret with total motile count and couple context.

Varicocele is dilation of the pampiniform venous plexus, classically left-sided because the left testicular vein drains into the left renal vein. It feels like a bag of worms and increases with standing or Valsalva. Varicocele can raise scrotal temperature and impair spermatogenesis. Repair is considered most strongly when there is a palpable varicocele, infertility, abnormal semen parameters, and a reproductive goal. A sudden new right-sided varicocele, noncompressible varicocele, or varicocele that does not reduce when supine raises concern for retroperitoneal or renal pathology obstructing venous drainage.

6. Acute Scrotum and Scrotal Masses

The acute scrotum is a surgical diagnostic domain because time matters. Testicular torsion occurs when the spermatic cord twists, obstructing venous return and then arterial inflow. Ischemic injury increases with time, so clinical suspicion should prompt urgent urologic consultation rather than waiting for perfect imaging. Doppler ultrasound can support the diagnosis when it does not delay care, but preserved or equivocal flow does not fully exclude intermittent or partial torsion.

Condition	Classic clues	Exam / imaging	Management concept
Testicular torsion	Sudden severe unilateral pain, nausea/vomiting, adolescent or young adult, high-riding testis	Absent cremasteric reflex; Doppler may show decreased flow	Urgent detorsion/orchiopexy; bilateral fixation because anatomic risk is often bilateral.
Epididymitis	Gradual posterior testicular pain, dysuria, urethral discharge, fever	Tender epididymis; Doppler often increased flow	Antibiotics by age/risk; STI testing; scrotal support/NSAIDs.
Torsion of appendix testis	Prepubertal boy, less toxic, localized upper pole pain	Blue dot sign sometimes; normal intratesticular flow	Supportive if torsion excluded.
Incarcerated hernia	Groin/scrotal mass, obstructive symptoms, pain	Mass may not transilluminate; bowel sounds possible	Urgent surgical evaluation if incarcerated/strangulated.
Testicular tumor with hemorrhage	Mass with acute pain from bleeding/infarct	Intratesticular lesion on ultrasound	Tumor markers and urology.

A scrotal mass is evaluated by location and transillumination. Intratesticular masses are malignant until proven otherwise; extratesticular masses are more often benign. Hydrocele is a fluid collection around the testis and usually transilluminates. Spermatocele is a cystic epididymal mass that may transilluminate and is typically benign. Varicocele feels like dilated veins and changes with position. Inguinal hernia may enlarge with Valsalva and can extend into the scrotum. Ultrasound distinguishes intratesticular solid lesions from extratesticular cystic or vascular lesions.

Mass	Location	Pain	Transillumination	Key association
Hydrocele	Fluid around testis	Usually painless	Yes	Can be idiopathic, reactive, traumatic, infectious, or tumor-associated.
Spermatocele	Epididymal cyst	Painless	Often yes	Benign sperm-containing cyst.
Varicocele	Pampiniform plexus	Dull ache/heaviness	No	Left-sided; infertility; does not reduce supine -> consider obstruction.
Testicular cancer	Intratesticular	Often painless	No	Firm mass; tumor markers; orchiectomy through inguinal approach.
Inguinal hernia	Can enter scrotum	Variable	No	May have bowel symptoms; incarcerated hernia is urgent.

7. Epididymitis, Orchitis, Urethritis, and Male STIs

Male reproductive infections are best organized by anatomic syndrome: urethritis causes dysuria and urethral discharge; epididymitis causes posterior testicular pain and swelling; orchitis causes testicular inflammation; prostatitis causes pelvic/perineal pain and urinary symptoms; genital ulcer syndromes cause penile lesions. The same pathogen can create different syndromes depending on anatomic spread. Chlamydia trachomatis and Neisseria gonorrhoeae are key causes in sexually active younger men, whereas enteric gram-negative organisms are more common in older men, men with urinary tract instrumentation, bladder outlet obstruction, or insertive anal exposure depending on context.

Syndrome	Typical clues	Likely organisms	Evaluation / treatment logic
Nongonococcal urethritis	Dysuria, mucoid discharge, urethral irritation	Chlamydia, Mycoplasma genitalium, Trichomonas in some settings	NAAT for GC/CT; doxycycline-based treatment for chlamydia/NGU; partner management.
Gonococcal urethritis	Purulent discharge, dysuria	Neisseria gonorrhoeae often with chlamydia coinfection	Ceftriaxone-based therapy; add doxycycline if chlamydia not excluded.
Epididymitis age <35 or STI risk	Unilateral epididymal pain + dysuria/discharge	GC/CT	Treat STI pathogens; abstain until treated; partner treatment.
Epididymitis age >35 / urinary risk	Pain with urinary symptoms, BPH/instrumentation	Enteric gram-negatives such as E. coli	Urine culture; fluoroquinolone may be used when gonorrhea not suspected and local risk allows.

Syndrome	Typical clues	Likely organisms	Evaluation / treatment logic
Orchitis	Testicular swelling/pain, viral prodrome possible	Mumps virus, enteroviruses, STI extension	Supportive if viral; evaluate bacterial/STI causes when indicated.
Genital ulcers	Painful vesicles/ulcers or painless chancre	HSV, syphilis, chancroid rare, noninfectious mimics	HSV PCR/culture, syphilis serology; HIV testing; empiric treatment when suspicion high.

CDC STI Treatment Guidelines remain the practical anchor for urethritis, gonorrhea, chlamydia, syphilis, herpes, and epididymitis management. The treatment logic is syndrome-based plus pathogen testing: test for gonorrhea/chlamydia using NAAT, treat presumptively when follow-up is uncertain or clinical suspicion is high, and address partners to prevent reinfection. Persistent or recurrent urethritis after standard therapy should raise suspicion for *Mycoplasma genitalium*, trichomoniasis in selected populations, reinfection, nonadherence, or noninfectious urethral irritation.

High-yield relationship: chlamydial urethritis can ascend to epididymitis; epididymal inflammation can cause pain, swelling, and rarely obstruction; epididymal obstruction can contribute to male infertility.

8. Benign Prostatic Hyperplasia and Lower Urinary Tract Symptoms

Benign prostatic hyperplasia is nonmalignant enlargement of periurethral/transitional-zone prostate tissue that can narrow the prostatic urethra and create bladder outlet obstruction. Symptoms are divided into voiding symptoms, such as hesitancy, weak stream, straining, intermittency, and incomplete emptying, and storage symptoms, such as urgency, frequency, and nocturia. The degree of symptoms does not always match prostate size because smooth muscle tone, bladder contractility, detrusor overactivity, medications, and fluid habits all contribute.

Medication class	Mechanism	Best fit	Important cautions
Alpha-1 blockers: tamsulosin, alfuzosin, doxazosin, terazosin	Relax prostatic/bladder neck smooth muscle	Rapid symptom relief, smaller or uncertain prostate size	Orthostasis; ejaculatory changes; floppy iris syndrome risk with cataract surgery.
5-alpha-reductase inhibitors: finasteride, dutasteride	Shrink prostate by reducing DHT	Larger prostate, progression prevention, high PSA/prostate volume context	Takes months; sexual side effects; lowers PSA roughly by half after sustained use.
PDE5 inhibitor: tadalafil	Smooth muscle effects in LUTS/ED	BPH with erectile dysfunction	Avoid nitrates; headache/flushing; drug interactions.
Antimuscarinic or beta-3 agonist	Treat bladder storage symptoms	Persistent urgency/frequency after obstruction considered	Avoid or monitor if high postvoid residual/retention risk.
Surgery / procedures	Relieve obstruction mechanically	Refractory symptoms, retention, stones, renal dysfunction, recurrent infections, hematuria	Choice depends on size, anatomy, bleeding risk, and patient goals.

Evaluation begins with history, medication review, focused exam, urinalysis, and symptom burden. Antihistamines, decongestants, anticholinergics, opioids, alcohol, and diuretics can worsen lower urinary tract symptoms. Red flags include hematuria, recurrent urinary tract infection, urinary retention, renal insufficiency, bladder stones, neurologic deficits, weight loss, or bone pain. Postvoid residual, creatinine, PSA, imaging, cystoscopy, or urodynamics are not mandatory for every mild case but become important when complications, diagnostic uncertainty, or invasive treatment are considered.

9. Prostatitis and Chronic Pelvic Pain Syndrome

Prostatitis is a syndrome label, not a single disease. Acute bacterial prostatitis causes fever, chills, dysuria, urinary frequency, pelvic/perineal pain, and a tender boggy prostate. Prostatic massage should be avoided in acute bacterial prostatitis because manipulation can worsen bacteremia. Chronic bacterial prostatitis causes recurrent urinary tract infections with the same organism and may have pelvic discomfort. Chronic prostatitis/chronic pelvic pain syndrome is more common than chronic bacterial prostatitis and involves pelvic pain, urinary symptoms, sexual dysfunction, central sensitization, pelvic floor dysfunction, inflammation, and psychosocial factors without consistent bacterial infection.

Syndrome	Clinical pattern	Testing	Treatment logic
Acute bacterial prostatitis	Fever, chills, dysuria, retention, perineal pain, tender prostate	Urinalysis/culture; blood cultures if toxic; avoid massage	Antibiotics with prostatic penetration; admit if septic/retention/unable oral.
Chronic bacterial prostatitis	Recurrent UTIs, pelvic discomfort	Urine culture; localization tests rarely used in primary care	Longer antibiotic course; evaluate obstruction/stones if recurrent.
Chronic pelvic pain syndrome	Pelvic/perineal pain >3 months, urinary/sexual symptoms, negative cultures	Exclude infection, malignancy, stones, neurologic disease when indicated	Multimodal: alpha blocker in selected, NSAIDs, pelvic floor PT, neuromodulators, psychosocial care.
Asymptomatic inflammatory prostatitis	Inflammatory findings without symptoms	Incidental	Usually no treatment unless specific fertility/PSA context.

10. Prostate Cancer and PSA Screening

Prostate cancer commonly arises in the peripheral zone and may be asymptomatic for years. Local disease may be found through PSA-based screening or abnormal digital rectal examination. Advanced disease can present with bone pain, weight loss, anemia, urinary obstruction, renal dysfunction, or pathologic fracture. PSA is organ-specific rather than cancer-specific: it can rise with cancer, BPH, prostatitis, urinary retention, recent instrumentation, and ejaculation. Therefore, PSA interpretation is probabilistic and should account for age, risk, medications, prostate size, trend, and clinical context.

Screening is a shared decision-making topic rather than a universal automatic test. USPSTF guidance supports individualized PSA screening decisions for men ages 55 to 69 and recommends against routine PSA-based screening in men 70 years or older. Men with higher risk, including Black men and men with a family history or germline cancer syndromes, may reasonably discuss screening earlier with clinicians under specialty guidance.

Finding	Interpretation	Clinical implication
Elevated PSA	Cancer possible but not specific	Repeat/confirm when appropriate; evaluate infection/retention; risk calculators/MRI/biopsy decisions.
Rapid PSA rise	Higher concern, but context matters	Assess prostatitis/instrumentation and refer when persistent.
Abnormal DRE	Nodule, asymmetry, induration	Urology evaluation even if PSA is not dramatically elevated.
5-alpha-reductase inhibitor use	PSA may fall about 50% after sustained therapy	Interpret PSA with medication adjustment and trend.
Bone pain + high PSA	Possible metastatic prostate cancer	Imaging and oncology/urology evaluation.

Prostate cancer diagnosis is made by biopsy, increasingly informed by multiparametric MRI and risk-stratified pathways. Risk stratification uses PSA, Gleason Grade Group, tumor stage, number/percentage of positive cores, imaging, and genomic tools in selected cases. Management ranges from active surveillance for low-risk disease to prostatectomy, radiation therapy, androgen deprivation therapy, androgen receptor pathway inhibitors, chemotherapy, radiopharmaceuticals, or palliative care depending on risk and stage.

Graph-RAG relationship: androgen receptor signaling drives many prostate cancers; androgen deprivation lowers testosterone signaling; castration-resistant prostate cancer progresses despite castrate testosterone levels through androgen receptor amplification, alternative signaling, or ligand-independent activation.

11. Testicular Cancer

Testicular cancer is the most important diagnosis behind a painless firm intratesticular mass in a young man. Germ cell tumors are the major category and are divided into seminoma and nonseminomatous germ cell tumors. Cryptorchidism increases risk, and orchiopexy reduces but does not eliminate that risk. Ultrasound is the key first imaging test for a testicular mass. Serum tumor markers - alpha-fetoprotein, beta-hCG, and LDH - support diagnosis, staging, risk stratification, and surveillance, but normal markers do not exclude cancer.

Tumor / marker pattern	Typical features	High-yield relationships
Seminoma	Often painless mass; radiosensitive/chemosensitive; may produce beta-hCG in some cases	AFP should not be elevated in pure seminoma; elevated AFP implies nonseminomatous component.
Embryonal carcinoma	Aggressive nonseminomatous tumor	May elevate AFP and/or beta-hCG; hemorrhage/necrosis more common.
Yolk sac tumor	Most common testicular tumor in young children	AFP often elevated.
Choriocarcinoma	Rare, aggressive, hematogenous spread	Very high beta-hCG; gynecomastia possible; lung/brain mets.
Teratoma	Mature or immature tissue elements	In adults considered malignant potential; chemo-resistant elements may require surgery.
LDH	Nonspecific tumor burden marker	Useful for staging/prognosis and surveillance trend.

Biopsy through the scrotum is avoided because it can disrupt lymphatic drainage and complicate management. Radical inguinal orchiectomy is the diagnostic and therapeutic route for suspicious solid intratesticular tumors. Testicular lymphatic drainage is retroperitoneal, not inguinal, unless prior scrotal surgery has altered drainage. Survivorship issues include fertility preservation, testosterone deficiency, contralateral testicular cancer risk, cardiovascular/metabolic effects of therapy, neuropathy, hearing loss, and psychosocial health.

12. Penile, Breast, and Other Male Reproductive Conditions

Penile cancer is uncommon in the United States but is associated with HPV infection, phimosis, chronic inflammation, smoking, and lack of neonatal circumcision. Persistent penile ulcers, masses, bleeding lesions, or nonhealing plaques require examination and biopsy rather than repeated empiric treatment. Condyloma acuminata are HPV-related warts and are distinct from invasive penile cancer, but HPV vaccination and safer sex practices reduce HPV-related disease burden.

Male breast cancer is rare but should be considered with a unilateral firm breast mass, nipple retraction, nipple discharge, skin dimpling, axillary adenopathy, Klinefelter syndrome, BRCA2 mutation, chest radiation history, or strong family history. Gynecomastia is usually concentric glandular tissue beneath the areola, whereas malignancy is more likely eccentric, hard, fixed, or associated with skin/nipple findings.

Condition	Core mechanism	Clinical clue	Management concept
Gynecomastia	Increased estrogen effect relative to androgen effect	Tender subareolar glandular tissue; puberty/aging/meds/liver disease/hypogonadism	Medication review, endocrine/testicular evaluation when persistent or suspicious.
Peyronie disease	Fibrotic plaque of tunica albuginea	Penile curvature, painful erections, erectile difficulty	Urology if bothersome, severe curvature, or sexual dysfunction.
Priapism	Prolonged erection; ischemic type is compartment syndrome	Painful rigid corpora >4 hours	Emergency aspiration/phenylephrine; sickle cell and medication evaluation.
Phimosis	Cannot retract foreskin	Balanitis, hygiene difficulty, urinary issues	Topical steroids/stretching or circumcision depending severity.
Paraphimosis	Retracted foreskin trapped behind glans	Painful swollen glans/foreskin	Urgent reduction to prevent ischemia.

13. Integrated Diagnostic Algorithms

Algorithm: acute testicular pain. First ask whether torsion is clinically plausible based on sudden onset, severe pain, nausea/vomiting, high-riding or transverse testis, and absent cremasteric reflex. If yes, call urology immediately; ultrasound should not delay intervention. If torsion is unlikely or excluded, evaluate epididymitis, torsion appendix testis, hernia, trauma, tumor, and referred pain from renal colic. Urinalysis, urine culture, and GC/CT NAAT are selected based on age and risk.

Algorithm: erectile dysfunction. Establish onset, severity, nocturnal/morning erections, libido, ejaculation, relationship context, cardiovascular symptoms, diabetes/HTN/smoking status, pelvic surgery/radiation, neurologic disease, and medication exposure. Screen for cardiometabolic disease and measure morning testosterone when low libido, infertility, testicular atrophy, osteoporosis, anemia, or other hypogonadal clues exist. Treat reversible causes, optimize vascular risk, use PDE5 inhibitors when safe, and refer for refractory disease or complex endocrine/fertility issues.

Algorithm: male infertility. Evaluate both partners. Obtain at least one semen analysis and repeat if abnormal. If semen volume is low, consider collection issue, retrograde ejaculation, androgen deficiency, or obstruction. If azoospermia is present, use testicular size, FSH, LH, testosterone, semen volume/pH, vas deferens exam, and history to separate obstructive from nonobstructive azoospermia. Avoid exogenous testosterone in men desiring fertility.

Algorithm: lower urinary tract symptoms. Assess severity and bother, urinalysis, medication contributors, neurologic disease, hematuria, retention, infection, renal dysfunction, and cancer symptoms. Mild symptoms can be managed with watchful waiting, lifestyle changes, and medication review. More bothersome voiding symptoms often respond to alpha blockers; large prostate/progression risk supports 5-alpha-reductase inhibitor discussion. Complications or diagnostic uncertainty require urologic evaluation.

14. Graph-RAG Relationship Summary

Source concept	Relationship	Target concept	Why the edge matters
Exogenous testosterone	suppresses	LH/FSH	Negative feedback lowers gonadotropin secretion.
Low LH/FSH	reduces	intratesticular testosterone	Intratesticular testosterone is required for sperm production.
Low intratesticular testosterone	causes	impaired spermatogenesis	Explains why testosterone therapy can worsen fertility.
Varicocele	increases	scrotal heat / venous stasis	Heat and oxidative stress can impair semen parameters.
Testicular torsion	causes	testicular ischemia	Twisting spermatic cord obstructs blood flow; time-sensitive emergency.
Chlamydia/gonorrhea	can cause	urethritis and epididymitis	Ascending infection produces pain, discharge, infertility risk.
BPH	causes	bladder outlet obstruction	Transitional-zone enlargement narrows prostatic urethra.
5-alpha-reductase inhibitors	lower	DHT and PSA	Shrink prostate over months; PSA interpretation changes.
Peripheral-zone prostate cancer	raises	PSA variably	PSA is nonspecific; cancer screening uses shared decision-making.
Cryptorchidism	increases risk of	testicular cancer and infertility	Temperature and abnormal testicular development affect germ cells.
beta-hCG secreting germ cell tumor	can cause	gynecomastia	hCG stimulates Leydig cells and increases estrogenic effect.
Diabetes mellitus	causes	ED and retrograde ejaculation	Vascular and autonomic neuropathy mechanisms.

15. High-Yield Review Tables

Diagnosis	Most useful clue	First-line / next step concept
Testicular torsion	Sudden pain, high-riding testis, absent cremasteric reflex	Urgent urology; do not delay for imaging when suspicion high.
Epididymitis	Posterior testicular tenderness + urinary/STI symptoms	NAAT/urine testing; antibiotics by organism risk.
Varicocele	Bag of worms; left-sided; worse standing	Semen analysis if infertility; repair when palpable + infertility + abnormal semen.
Hydrocele	Painless fluid collection that transilluminates	Ultrasound if unable to examine testis or concern for secondary cause.
BPH	Weak stream, nocturia, hesitancy	Alpha blocker for rapid symptoms; 5-ARI for large prostate/progression.
Acute bacterial prostatitis	Fever + tender boggy prostate	Urine culture + antibiotics; avoid prostate massage.
Prostate cancer	Elevated PSA or abnormal DRE; often asymptomatic	Shared screening decision; MRI/biopsy pathway when indicated.
Testicular cancer	Painless firm intratesticular mass	Ultrasound, AFP/beta-hCG/LDH, radical inguinal orchiectomy.
Hypogonadism	Symptoms + repeatedly low AM testosterone	Determine primary vs secondary with LH/FSH; fertility-aware treatment.

References and guideline anchors

American Urological Association. Testosterone Deficiency Guideline. AUA guideline statements include using total testosterone below 300 ng/dL as a reasonable diagnostic cut-off when symptoms/signs are present.

American Urological Association / American Society for Reproductive Medicine. Diagnosis and Treatment of Infertility in Men: AUA/ASRM Guideline, amended 2024.

Centers for Disease Control and Prevention. Sexually Transmitted Infections Treatment Guidelines, 2021, with current online updates and wall-chart summaries.

U.S. Preventive Services Task Force. Prostate Cancer: Screening recommendation statement; individualized PSA screening decisions for men 55-69 and recommendation against routine screening at age 70 or older.

Standard internal medicine, urology, endocrinology, infectious disease, and oncology teaching frameworks were used for original synthesis and organization.